

[153]

SEAT No. _____



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SARDAR PATEL UNIVERSITY
Master of Science (Information Technology)-Third Semester
PS03CINT33 : Computer Graphics

25th November, 2021, ~~Thursday~~

Time: 3.00 pm to 5.00 pm

Marks: 70

Q1.A Choose the most appropriate option for the following

[8]

- i. CAD stands for _____
 (A) Computer Aided Design (B) Common Aided Design
 (C) Common Application Design (D) Computer Application Design
- ii. _____ is produced by capping the connection between the two segments with a circular boundary.
 (A) Bevel Join (B) Square Join (C) Round Join (D) Miter Join
- iii. Which of the following transformation alter the size of an object?
 (A) Translation (B) Rotation (C) Scaling (D) Reflection
- iv. The 3D object which cannot retain its shape is called _____.
 (A) Spline object (B) Curved object (C) Blobby object (D) None of given
- v. The process of removing unwanted sounds that crept in, during the audio Recordings is known as _____.
 (A) Dithering (B) Trimming (C) Noise removal (D) None of given
- vi. Flash is _____ authoring tool.
 (A) Time based (B) Frame based (C) Icon based (D) None of given
- vii. The video industry uses _____ color model.
 (A) CMY (B) YCbCr (C) RGB (D) None of given
- viii. The _____ file format does not allow to store an animation sequence.
 (A) GIF (B) TIFF (C) jpg (D) All of given

Q1.B Do as directed.

[8]

(a) Fill in the blanks:

- i. A _____ transformation produces a mirror image of an object.
- ii. A world coordinate area selected for display is called _____.
- iii. The full form of DDA is _____.
- iv. In 3D, coordinate value for each vertex of an object is stored in _____ table..
- v. The data transfer rate of 48X CD ROM drive is _____ KB/sec.
- vi. The process of changing the RGB value of the color pixels that make up the image, to the closest matching color in predefined fixed color palette, using complex mathematical algorithms is known as _____.
- vii. The technique of applying 2D images over 3D wire frame models is called _____.
- viii. The authoring tool, Macromedia Director uses _____ authoring methodology.

(P.T.O)

(b) State whether each of the following statement is True or False: [8]

- i. The Sutherland-Hodgeman clipping algorithm is used for polygon clipping.
- ii. The mapping of a part of a world-coordinate scene to device coordinates is referred to a geometric transformation.
- iii. A circle is defined as the set of points that are all at a given fixed distance from a center position.
- iv. Any procedure that identifies those portions of a picture that are either inside or outside of a specified region of space is referred as clipping algorithm.
- v. Software that provides the necessary interface required for the interaction, between the operating system and the Multimedia hardware peripherals attached to the system are known as Device driver.
- vi. In Computer industry, CMY color model is used.
- vii. In human eyes, the photoreceptor cells cones are responsible for identification of colors.
- viii. Image restoration may be defined as an attempt to estimate the original image by applying effective inversion of the degrading phenomenon.

Q2. Answer in short (Attempt any SEVEN)

[14]

- i. Differentiate clearly Raster-Scan display and Random-scan display.
- ii. Explain briefly 2D translation transformation.
- iii. Write an equation of circle using polar coordinates r and θ .
- iv. Explain briefly the shear transformation.
- v. Define: Multimedia.
- vi. Explain briefly two main characteristics of multimedia text.
- vii. Calculate the size of audio file in bits, having 32 bit sound system, recording signals at 24 KHz in stereo recording for 20 seconds.
- viii. List out the types of audio cards with their usage
- ix. Write the main difference between Multimedia Programming and Multimedia authoring

Q3. Do as directed.

[8]

- [A] Write the steps to draw line using Bresenham's line drawing algorithm.
- [B] Assume that a polygon with vertices (5, 2), (15, 2), (5, 10) and (15, 10) is given. Find the new co-ordinates of polygon after performing the translation to the given polygon with respect to the translation vector (4, 2).

OR

Q3. Do as directed.

[8]

- [A] Explain briefly text clipping algorithms.
- [B] Explain briefly any TWO character generation methods.

Q4. Do as directed.

[8]

- [A] Explain briefly the Parallel projection and Perspective projection.
- [B] List out the basic three dimensional geometric transformations. Explain any one of them in detail.

OR

Q4. Do as directed.

[8]

- [A] Write short note on 3-D viewing pipeline.
- [B] Write the steps for z-buffer algorithm, a visible surface detection method.

Q5. Do as directed.

[8]

- [A] Define Image digitization. Also explain briefly Image digitization process.
- [B] Smooth the pixel $g(2,2)$ which is underlined, of a following grayscale image having 25 pixels, in 3×3 domain using mean filter and median filter.

1	2	<u>2</u>	4	5
4	5	0	1	3
3	4	<u>6</u>	7	0
2	1	4	6	1
3	5	2	5	2

OR

Q5 Do as directed.

[8]

- [A] Write the steps to convert color value available in YCbCr color model to CMY color model.
- [B] List out the broad classification of Image processing & Analysis with their input and output. Also write the components of Image processing & Analysis system.

Q6. Do as directed.

[8]

- [A] List out the different facets of Multimedia. Also state different hardware required to handle different media in the computer.
- [B] Write the short note on multimedia project design and development.

OR

Q6. Do as directed.

[8]

- [A] Write at least 4 reasons for audio editing. Also explain briefly Digital recording process & Digital playback process.
- [B] Write short note on 3-dimensional animation development process.

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(3)

